Grade 9 Mathematics PT-2

Core (2) - Transfer + H1/H2/H3 + Semantic Reasoning Route

Hints support the attempt. The reasoning route explains the logic. The solution shows the finished mathematics.

QUESTION | CONCEPT | EASY / MEDIUM / HARD | CORE1 | SOURCE | EVIDENCE

DIFFICULTY

EASY = one direct idea or substitution. MEDIUM = several linked reasoning states. HARD = multi-stage modelling, state reuse or substantial algebra.

Difficulty controls route depth and page allocation.

REASONING ROLES

INTERPRET → understand the demand. REPRESENT / MODEL → turn it into mathematics. EXECUTE / TRANSFORM → carry out valid steps. COMPARE / INFER → conclude. VERIFY → independently check.

Each route state can link directly to H1/H2/H3 and to Core (1).

Use: Attempt first. Reveal H1, then H2, then H3 only if needed. Read the reasoning route after attempting. Read the solution last.

PT-2 Q1 | Euclid: axiom or postulate?

Q1 | EUCLID FOUNDATIONS | EASY | Core1 C1 | SRC PT2-Q1 | EVID SW-Q1

QUESTION / ATTEMPT

Which of the following statements is NOT an axiom of Euclid?

a) Things which are equal to the same thing are equal to one another.

b) If equals are added to equals, the wholes are equal.

c) A circle can be drawn with any centre and any radius.

d) If equals are subtracted from equals, the remainders are equal.

Choice: ____

Reason: ______________________________

H1 clue / structure

AXIOM = general truth
POSTULATE = geometry-specific starting rule

H2 representation / method

Classify each option as a general equality rule or a geometry construction rule.

H3 first executable step

The statement 'A circle can be drawn...' is a construction postulate.

REASONING ROUTE

1 INTERPRET **Read the word NOT** Core1 C1 H1
You need the one statement that is not an axiom.

2 COMPARE **Classify** H2
a, b and d are general equality truths.

3 INFER **Select** H3
c is the geometry construction postulate.

SOLUTION

Correct option: (c).

A circle can be drawn with any centre and any radius is Euclid's postulate, not a general axiom. The other three statements are equality axioms.

CHECK A geometry construction rule is a postulate; equality rules apply more generally.

PT-2 Q2 | Distance between two points

Q2 | COORDINATE DISTANCE | EASY | Core1 D1 | SRC PT2-Q2 | EVID SW-Q2

QUESTION / ATTEMPT

Distance between two points A(x1,y1) and B(x2,y2) is given by:

a) √[(x2-y2)^2 + (x1-y1)^2]

b) √[(x2+y2)^2 - (x1+y1)^2]

c) √[(x2+x1)^2 - (y2+y1)^2]

d) [(x2-x1)^2 + (y2-y1)^2]^(1/2)

Formula: ____

Choice: ____

H1 clue / structure

Δx = x2 - x1
Δy = y2 - y1

H2 representation / method

d^2 = (Δx)^2 + (Δy)^2
Use Pythagoras.

H3 first executable step

d = √[(x2-x1)^2 + (y2-y1)^2]
Match this with the options.

REASONING ROUTE

1. REPRESENT **Horizontal change** — Core1 D1 | H1
   Δx=x2-x1

2. REPRESENT **Vertical change**
   Δy=y2-y1

3. INFER **Pythagoras** — H2
   d=√[(Δx)^2+(Δy)^2]

SOLUTION

Correct option: (d).
d = √[(x2-x1)^2 + (y2-y1)^2]

CHECK If A and B are the same point, both differences are zero and the distance is zero.

PT-2 Q3 | Quadrant after a sign transformation

Q3 · QUADRANTS · SIGNS · MEDIUM · Core1 B2 · SRC PT2-Q3 · EVID SW-Q3

QUESTION / ATTEMPT

If x < 0 and y < 0, then the point (-x, y) lies in:

a) I quadrant

b) II quadrant

c) III quadrant

d) IV quadrant

Sign of -x: ____

Sign of y: ____

Quadrant: ____

H1 clue / structure

x=-4, y=-2 ⇒ (-x,y)=(4,-2)

H2 representation / method

(+,-) means right of the y-axis and below the x-axis.

H3 first executable step

(+,-) ⇒ Quadrant IV

REASONING ROUTE

1. INTERPRET **Choose legal sample** — Core1 B2 · H1
   x=-4, y=-2
2. TRANSFORM **Transform point**
   (-x,y)=(4,-2)
3. COMPARE **Read signs** — H2
   (+,-)
4. INFER **Map quadrant** — H3
   (+,-)→IV

SOLUTION

Correct option: (d), IV quadrant.

Because x<0, -x>0 while y remains negative.
(-x,y) has signs (+,-) ⇒ Quadrant IV

CHECK Classify the transformed point (-x,y), not the original signs of (x,y).

PT-2 Q4 | Find the y-intercept

Q4 INTERCEPT LINEAR EQUATION EASY Core1 E1 SRC PT2-Q4 EVID SW-Q4

QUESTION / ATTEMPT

The graph of line $5x + 3y = 4$ cuts the y-axis at the point:

a) (4/3, 0) **b)** (0, 4/3)

c) (4/5, 0) **d)** (5/4, 0)

Set x = ___

y = ___

Point: _________

H1 clue / structure

On the y-axis: $x = 0$

H2 representation / method

$5(0) + 3y = 4$

H3 first executable step

$3y=4 \Rightarrow y=4/3$

REASONING ROUTE

1 INTERPRET **Use axis condition** Core1 E1 H1

$x=0$

2 EXECUTE **Substitute** H2

$3y=4$

3 INTERPRET RESULT **Form point** H3

$(0,4/3)$

SOLUTION

Correct option: (b), (0, 4/3).

$5(0)+3y=4$

$y=4/3$

CHECK Every point on the y-axis has x-coordinate 0.

PT-2 Q5 | Line with slope 2/3 through (4,-1)

Q5 | SLOPE | Y=MX+C | MEDIUM | Core1 E2 | SRC PT2-Q5 | EVID SW-Q5

QUESTION / ATTEMPT

Which is an equation of the line with coefficient of x as 2/3 and that passes through the point (4,-1)?

a) y = (-1/4)x + 2/3

b) y = -4x + 2/3

c) y = (2/3)x - 5/3

d) y = (2/3)x - 11/3

m = ____

Substitute point: ________________________

c = ____

H1 clue / structure

y = mx + c, m = 2/3

H2 representation / method

-1 = (2/3)(4) + c
-1 = 8/3 + c

H3 first executable step

c = -11/3
y=(2/3)x - 11/3

REASONING ROUTE

1. INTERPRET **Read slope** (Core1 E2, H1)
   m=2/3
2. REPRESENT **Use line form**
   y=mx+c
3. EXECUTE **Substitute point** (H2)
   -1=8/3+c
4. INFER **Solve c** (H3)
   c=-11/3

SOLUTION

Correct option: (d), y=(2/3)x - 11/3.

-1=(2/3)(4)+c=8/3+c
c=-11/3

CHECK At x=4, y=8/3-11/3=-1, so the given point lies on the line.

PT-2 Q6 | Lines through six points

Q6 | PAIR COUNTING | MEDIUM | Core1 F1 | SRC PT2-Q6 | EVID SW-Q6

QUESTION / ATTEMPT

Given 6 points such that no three of them are collinear, the number of lines that can be drawn through any two points is:

a) 120

b) 6

c) 15

d) none of these

Counting idea: ____________________________

Calculation: ____________________________

H1 clue / structure

One distinct line needs one unordered pair of points.

H2 representation / method

$C(6,2) = 6\times5 / 2$

H3 first executable step

$6\times5/2 = 15$

REASONING ROUTE

1 INTERPRET **Identify object** Core1 F1 H1

A line is determined by a pair.

2 REPRESENT **Count pairs** H2

$C(6,2)$

3 EXECUTE **Evaluate** H3

15

SOLUTION

Correct option: (c), 15.

$C(6,2)=6\times5/2=15$

CHECK AB and BA are the same line, so order is not counted twice.

PT-2 Q7 | Check collinearity without plotting

Q7 · SLOPE · COLLINEARITY · MEDIUM · Core1 D2 · SRC PT2-Q7 · EVID SW-Q7

QUESTION / ATTEMPT

Without plotting check whether the points A(-2,-2), B(8,2) and C(3,0) are collinear.

m_AB = ____________________

m_AC = ____________________

Conclusion: __________________

H1 clue / structure

Test: m_AB ?= m_AC

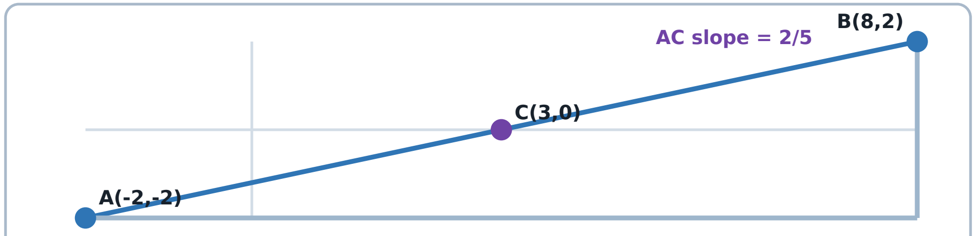
H2 representation / method

m_AB=(2+2)/(8+2)=4/10
m_AC=(0+2)/(3+2)=2/5

H3 first executable step

4/10 = 2/5

Equal slopes ⇒ same straight line.



REASONING ROUTE

1. INTERPRET **Identify the test** (Core1 D2)
Collinear means the three points lie on one straight line.
2. VERIFY **Check copied data**
A(-2,-2), B(8,2), C(3,0)
3. REPRESENT **Use one common anchor** (H1)
Compare m_AB and m_AC
4. EXECUTE **Direction A→B** (H2)
m_AB = 4/10 = 2/5
5. EXECUTE **Direction A→C**
m_AC = 2/5
6. COMPARE **Compare directions** (H3)
m_AB = m_AC
7. INFER **Interpret geometrically**
Same anchor A + same slope means B and C lie on the same straight line through A.

SOLUTION

The three points are collinear.

m_AB=(2-(-2))/(8-(-2))=4/10=2/5
m_AC=(0-(-2))/(3-(-2))=2/5

CHECK Equal slopes from the same anchor point A mean B and C lie on the same straight line through A.

PT-2 Q8 | Euclid's Fifth Postulate and parallel lines

Q8 · PARALLEL LINES · EUCLID V · MEDIUM · Core1 C2 · SRC PT2-Q8 · EVID SW-Q8

QUESTION / ATTEMPT

Explain why Euclid's Fifth Postulate is considered the basis for understanding parallel lines. Support your answer with an example.

Condition: ______________________________

Explanation: ___________________________

Example: _______________________________

H1 clue / structure

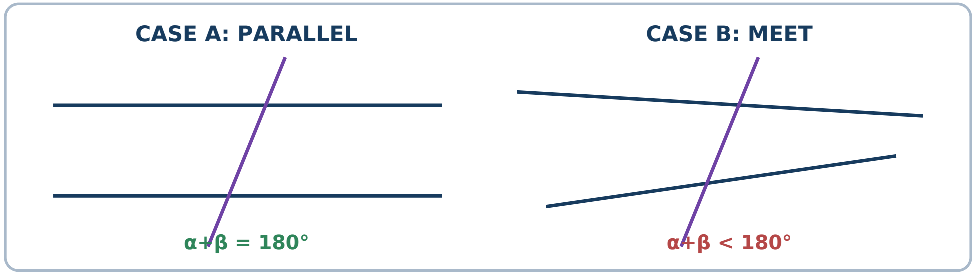
$\alpha + \beta < 180°$ ⇒ lines meet
$\alpha + \beta = 180°$ ⇒ parallel

H2 representation / method

Use same-side interior angles made by a transversal.

H3 first executable step

$70°+40°=110°<180°$
So those lines meet; they are not parallel.



REASONING ROUTE

1. INTERPRET **Identify the geometry** (Core1 C2)
   Two lines are cut by a transversal.
2. REPRESENT **Choose the relevant angles** (H2)
   Use the same-side interior angles α and β.
3. INFER **Meet condition** (H1)
   $\alpha+\beta < 180°$ ⇒ lines meet
4. CHECK CONDITION **Parallel boundary**
   $\alpha+\beta = 180°$ ⇒ parallel
5. EXECUTE **Test a numerical case** (H3)
   $70°+40° = 110° < 180°$
6. INTERPRET RESULT **Conclude**
   Those two lines meet, so they are not parallel.

SOLUTION

Euclid's Fifth Postulate links the interior-angle sum to whether two lines meet.

If same-side interior angles sum to less than 180°, the lines meet on that side.
If the sum is 180°, the lines are parallel.
Example: $70°+40°=110°<180°$ ⇒ the lines meet.

CHECK Do not reverse the conclusion: a sum below 180° means meeting, not parallel.

PT-2 Q9 | Equal x and y in a parameter equation

Q9 · PARAMETER · INFORMATION SUFFICIENCY · MEDIUM · Core1 F5 · SRC PT2-Q9 · EVID SW-Q9

QUESTION / ATTEMPT

For what value of c, the linear equation $2x + cy = 8$ has equal values of x and y for its solution?

Substitute x=y: ________________________

Can c be unique? ________________________

H1 clue / structure

$x = y$

H2 representation / method

$2x + cx = 8$

$x(2+c)=8$

H3 first executable step

$c = 8/x - 2$

No x is given, so c is not unique.

REASONING ROUTE

1. INTERPRET **Use condition** (Core1 F5, H1)
   $y=x$
2. REPRESENT **Substitute** (H2)
   $x(2+c)=8$
3. TRANSFORM **Isolate c** (H3)
   $c=8/x-2$
4. CHECK CONDITION **Check sufficiency**
   No value of x is supplied, so no unique c exists.

SOLUTION

As printed, the question is underdetermined.

$x(2+c)=8$

$c=8/x-2$

Different nonzero x values give different c values.

CHECK The learner's symbolic result is legitimate; this item should not be used as negative evidence.

PT-2 Q10 | Intersection first, then a new line

Q10 | SIMULTANEOUS EQUATIONS | LINE EQUATION | HARD | Core1 B1 + Core1 E3 | SRC PT2-Q10 | EVID SW-Q10

QUESTION / ATTEMPT

Find the equation of the straight line which passes through the point (2,-3) and the point of intersection of the lines $x+y+4=0$ and $3x-y-8=0$.

Intersection point: ______________________

Slope with (2,-3): ______________________

Final line: ___________________________

H1 clue / structure

$x+y=-4$
$3x-y=8$

H2 representation / method

Add ⇒ $4x=4$
$x=1$, $y=-5$

H3 first executable step

$m=[-3-(-5)]/(2-1)=2$
$y+5=2(x-1)$

REASONING ROUTE

1. REPRESENT **Normalize equations** — Core1 E3, H1
   $x+y=-4$; $3x-y=8$
2. EXECUTE **Eliminate y**
   $4x=4$
3. INTERPRET RESULT **Store intersection** — H2
   $P=(1,-5)$
4. REPRESENT **Pair with given point**
   $Q=(2,-3)$
5. EXECUTE **Find slope** — H3
   $m=2$
6. TRANSFORM **Build line**
   $y+5=2(x-1)$
7. VERIFY **Simplify + verify**
   $y=2x-7$

SOLUTION

Required line: $y = 2x - 7$.

$x+y=-4$, $3x-y=8$ ⇒ $4x=4$ ⇒ $x=1$, $y=-5$
$m=(-3+5)/(2-1)=2$
$y+5=2(x-1)$ ⇒ $y=2x-7$

CHECK Both (1,-5) and (2,-3) satisfy $y=2x-7$.

PT-2 Q11 | Point on x-axis equidistant from two points

Q11 · DISTANCE EQUALITY · X-AXIS CONSTRAINT · HARD · Core1 F2 + Core1 B3 · SRC PT2-Q11 · EVID SW-Q11

QUESTION / ATTEMPT

Find the point on the x-axis, which is equidistant from the point (7,6) and (3,4).

Let P = _________

Distance equation: ______________________

Point: _______________________________

H1 clue / structure

P=(a,0)

Because P lies on the x-axis.

H2 representation / method

PA^2 = PB^2

H3 first executable step

(a-7)^2+36 = (a-3)^2+16

REASONING ROUTE

1. REPRESENT **Represent unknown** — Core1 F2, H1
   P=(a,0)
2. MODEL **Translate equidistant** — H2
   PA^2=PB^2
3. EXECUTE **Write distance equation** — H3
   (a-7)^2+36=(a-3)^2+16
4. TRANSFORM **Expand both sides** — Core1 B3
   a^2-14a+85=a^2-6a+25
5. EXECUTE **Solve**
   60=8a ⇒ a=15/2
6. VERIFY **State point**
   P=(15/2,0)

SOLUTION

The point is (15/2, 0).

(a-7)^2+36=(a-3)^2+16

-14a+85=-6a+25

a=15/2

CHECK Substituting a=15/2 gives equal squared distances to (7,6) and (3,4).

PT-2 Q12 | Third vertex of an equilateral triangle

Q12 | EQUILATERAL GEOMETRY | DISTANCE | HARD | Core1 B3 + Core1 F3 | SRC PT2-Q12 | EVID SW-Q12

QUESTION / ATTEMPT

If the two vertices of an equilateral triangle be (0,0), (3,√3), find the third vertex.

AB^2 = _____

Two equal-distance equations: _________________________

Third vertex/vertices: _________________

H1 **clue / structure**

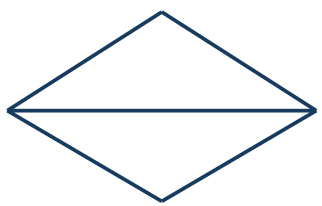
AB^2 = 3^2 + (√3)^2 = 12
AC^2 = BC^2 = 12

H2 **representation / method**

x^2+y^2=12
(x-3)^2+(y-√3)^2=12

H3 **first executable step**

(x-3)^2=x^2-6x+9
Not x^2-6x+81.



Same side length

Two mirror solutions

REASONING ROUTE

1. REPRESENT **Find side length** — Core1 F3, H1
AB^2=12
2. MODEL **Name third vertex**
C=(x,y)
3. REPRESENT **Equal distance to A** — H2
x^2+y^2=12
4. REPRESENT **Equal distance to B**
(x-3)^2+(y-√3)^2=12
5. TRANSFORM **Expand correctly** — Core1 B3, H3
(x-3)^2=x^2-6x+9
6. EXECUTE **Solve system**
Two mirror-image solutions
7. VERIFY **State both**
(0,2√3) or (3,-√3)

SOLUTION

Two valid third vertices: (0, 2√3) and (3, -√3).

x^2+y^2=12

(x-3)^2+(y-√3)^2=12

Solving the pair gives the two mirror-image third vertices.

CHECK Each candidate is at squared distance 12 from both given vertices.

PT-2 Q13 | Linear price-demand relationship

Q13 | SLOPE WITH UNITS | LINEAR MODEL | MEDIUM | Core1 E4 | SRC PT2-Q13 | EVID SW-Q13

QUESTION / ATTEMPT

The owner of a milk store finds that he can sell 980 litres milk each week at ₹14 per litre and 1220 litres each week at ₹16 per litre. Assuming a linear relationship between selling price and demand, how many litres could he sell weekly at ₹17 per litre?

Slope with units: ______________________

Prediction at ₹17: ______________________

H1 clue / structure

m = Δdemand / Δprice

H2 representation / method

m=(1220-980)/(16-14)=240/2

H3 first executable step

m=120 L per ₹
1220+120=1340 L

REASONING ROUTE

1. INTERPRET **Assign roles** — Core1 E4
   x=price, y=demand
2. REPRESENT **Compute vertical change**
   Δy=1220-980=240
3. EXECUTE **Compute horizontal change**
   Δx=16-14=2
4. INTERPRET RESULT **Slope with units** — H2
   m=240/2=120 L/₹
5. VERIFY **Extend to ₹17** — H3
   1220+120=1340 L

SOLUTION

Predicted weekly demand at ₹17/L: 1340 litres.

m=(1220-980)/(16-14)=120 L/₹
Demand at ₹17 = 1220 + 120 = 1340 L

CHECK This answer follows the linear relationship stated in the question, even though the real-world trend is unusual.

PT-2 Q14 | Boat and river current

Q14 | RATES | SIMULTANEOUS EQUATIONS | HARD | Core1 B1 + Core1 E3 + Core1 F4 | SRC PT2-Q14 | EVID SW-Q14

QUESTION / ATTEMPT

A boat travels 48 km downstream from A to B in 2 hours and returns upstream over the same distance in 3 hours. The boat's speed in still water is x km/h and the river current is y km/h. (i) Form the downstream equation. (ii) Form the upstream equation. (iii) Find x and y.

Downstream equation: ____________________

Upstream equation: ______________________

x = _____, y = _____

H1 clue / structure

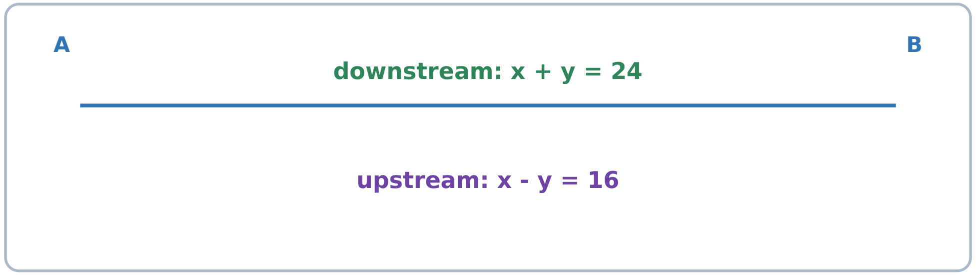
downstream speed = x + y
upstream speed = x - y

H2 representation / method

48/2 = 24 km/h
48/3 = 16 km/h

H3 first executable step

x+y=24
x-y=16



REASONING ROUTE

1. INTERPRET **Define variables** — Core1 F4
x=still-water speed; y=current speed

2. MODEL **Build downstream speed** — H1
x+y

3. EXECUTE **Use first trip** — H2
x+y=48/2=24

4. MODEL **Build upstream speed**
x-y

5. EXECUTE **Use return trip** — H3
x-y=48/3=16

6. TRANSFORM **Solve system** — Core1 E3
2x=40 ⇒ x=20

7. VERIFY **Recover + verify** — Core1 B1
y=4

SOLUTION

Boat speed in still water = 20 km/h; river current = 4 km/h.

x+y=24
x-y=16
Add: 2x=40 ⇒ x=20
Then y=4

CHECK 20+4=24 gives 48 km in 2 h; 20-4=16 gives 48 km in 3 h.

PT-2 Quick Handout

MEMORY AID | USE AFTER CORE1 | PT2 LINKS

ALGEBRA / EQUATIONS

- Legal move: apply the same operation to both sides.
- (a-b)^2 = a^2 - 2ab + b^2
- Eliminate one variable, then substitute back.
- Verify the result in the original equations.

COORDINATES / LINES

- d = √[(x2-x1)^2 + (y2-y1)^2]
- m = (y2-y1)/(x2-x1)
- y-axis means x=0
- Line form: y = mx + c

GEOMETRY / EUCLID

- Collinear = same straight line
- Same-side interior angles = 180° → parallel
- Same-side interior angles < 180° → lines meet
- Pairs from n points = n(n-1)/2

MODELLING / HABITS

- speed = distance/time
- downstream = x+y ; upstream = x-y
- Read → Copy → Point back → Check
- If information is insufficient, say so.